

Group 2

Requirements and Process Modeling

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Elicitation Progress

No stakeholder elicitation has been completed. The requirements below are provisional team proposals based on the project brief and discussions, not confirmed stakeholder findings.

The remaining work is to apply the selected techniques, record findings for each stakeholder, resolve conflicting needs, and revise the requirements. Optional extensions will be considered after the core incident workflow is working.

1. Stakeholders and Elicitation

The following techniques have been selected. The focus column lists questions to investigate, not confirmed interview findings.

No stakeholder elicitation has been completed. Before submission, each status must be refined to "In progress" or "Not yet started" based on the actual work done.

Stakeholder	Selected technique	Status	Focus and reason
On-Call Engineer / Resolver	Interview	Not completed	Understand alert noise, investigation needs, and handoffs from the responder's experience.
Incident Commander	Interview	Not completed	Clarify coordination, ownership, and decisions during an incident.
Service Owner	Interview	Not completed	Understand service impact, response targets, and escalation expectations.
Scribe	Observation and interview	Not completed	Examine how actions and decisions are recorded and corrected.
Communications Lead	Interview	Not completed	Clarify approval of updates and separation of internal and public information.
Engineering Manager	Interview and survey	Not completed	Compare reporting needs and expectations about team-level metrics.
Customer Support Lead	Interview	Not completed	Understand early-warning needs and how customer reports reach responders.
Platform Administrator	Interview and document analysis	Not completed	Clarify permissions, configuration, and integration setup.
Compliance / Audit Function	Document analysis	Not completed	Identify relevant record-keeping needs where no representative is available.
End Customer / End User	Survey and document analysis	Not completed	Gather expectations about outage updates and review status-page conventions.
External Systems	API documentation analysis	Not completed	Identify required alert fields, authentication, duplicate handling, and delivery behavior.

External systems are non-human actors, such as monitoring tools, GitHub Actions, and notification channels. Their interface documents will be used to confirm integration requirements.

Conflicts to discuss include alert frequency versus alert noise, reporting versus privacy, and automation versus human approval. Their resolutions are not yet claimed as stakeholder agreements.

2. Functional Requirements

These requirements describe the proposed system behavior. MoSCoW priorities are provisional: Must is core, Should is an important extension, and Could is an optional addition. Automatic remediation is Won't-have for the initial version. A sub-item's priority overrides its parent priority.

Source labels show the current basis and the stakeholder or technique planned for validation. "Team proposal" means an assumption to check, not a completed interview or document-analysis finding. Sub-items inherit their parent's source unless stated otherwise.

FR-01: Login and Access Control

Priority: Must

Source: Team proposal; validate with Platform Administrator; interview and document analysis.

- FR-01.1 The system shall allow users to log in and log out.
- FR-01.2 Users shall only view information and perform actions allowed by their permissions.
- FR-01.3 Application roles, such as Admin and Viewer, shall be separate from incident responsibilities, such as Commander and Scribe.

FR-02: Incident Creation

Priority: Must

Source: Team proposal; validate with Resolver and Incident Commander; interviews.

- FR-02.1 Authorized users shall be able to create an incident with a title, description, affected service, and severity.
- FR-02.2 The system shall generate a unique ID, record the reporter and creation time, and set the initial status to OPEN.

FR-03: Alert Intake and Grouping

Priority: Must

Source: Team proposal; validate with External Systems and Resolver; API document analysis and interview.

- FR-03.1 The system shall check the source and required fields of incoming alerts and reject invalid requests.
- FR-03.2 Receiving the same event again shall not create another alert or incident.
- FR-03.3 A new alert related to an active incident shall be linked to it using configured matching rules.
- FR-03.4 If no matching incident is found, the system shall create a new incident.

FR-04: Severity and Priority

Priority: Must

Source: Team proposal; validate with Service Owner and Incident Commander; interviews.

- FR-04.1 The system shall support SEV1, SEV2, SEV3, and SEV4.
- FR-04.2 Authorized users shall be able to change severity, with the reason recorded.
- FR-04.3 The incident board shall allow sorting by severity and time remaining before an SLA deadline.

FR-05: Incident Assignment

Priority: Must

Source: Team proposal; validate with Resolver and Platform Administrator; interviews.

- FR-05.1 The system shall assign a new incident to the service's on-call responder or configured fallback contact.
- FR-05.2 Authorized users shall be able to reassign an incident to another person or team.
- FR-05.3 The system shall display the current owner and record who changed the assignment, when, and from whom to whom.

FR-06: Incident Status

Priority: Must

Source: Team proposal; validate with Incident Commander and Resolver; interviews.

- FR-06.1 The system shall support OPEN, ACKNOWLEDGED, INVESTIGATING, IDENTIFIED, MONITORING, RESOLVED, and CLOSED.
- FR-06.2 Only authorized users or configured system actions shall be able to change the status.
- FR-06.3 The system shall allow only approved transitions and record each accepted change with its actor and time.
- FR-06.4 An invalid transition shall be rejected without changing the incident.

FR-07: Notifications and Escalation

Priority: Must

Source: Team proposal; validate with Resolver and Service Owner; interviews.

- FR-07.1 The system shall notify the assigned responder in the application.
- FR-07.2 **Could:** Email or a Slack webhook may be added as an external notification channel.
- FR-07.3 If an incident remains unacknowledged at the deadline and another escalation level exists, the system shall notify that level.
- FR-07.4 Accepted acknowledgement shall cancel pending acknowledgement escalations.
- FR-07.5 If the incident remains unacknowledged after all levels are exhausted, the system shall notify the fallback contact and leave it open.
- FR-07.6 Escalation deadlines shall be stored persistently and re-evaluated after restart.
- FR-07.7 Before escalating, the system shall check the stored deadline and current incident record. Only an OPEN incident with no accepted acknowledgement shall be eligible.
- FR-07.8 Notification attempts, failures, and escalation events shall be recorded.
- FR-07.9 **Could:** If external delivery is included, failures shall be visible for retry.

FR-08: Incident Timeline

Priority: Must

Source: Team proposal; validate with Scribe; observation and interview.

- FR-08.1 The system shall keep a timeline of alerts, status changes, assignments, comments, notifications, escalations, and response actions.
- FR-08.2 Each entry shall identify what happened, when, and who or what performed it.
- FR-08.3 Corrections shall be added as new entries linked to the original entry. The original shall remain unchanged.

FR-09: Collaboration

Priority: Must

Source: Team proposal; validate with Resolver and Scribe; interviews.

- FR-09.1 Authorized users shall be able to add comments, investigation findings, and updates to an incident.
- FR-09.2 These contributions shall remain available in the incident record and follow the incident's access permissions.

FR-10: Root Cause and Resolution

Priority: Must

Source: Team proposal; validate with Resolver and Service Owner; interviews.

- FR-10.1 Responders shall be able to record suspected or confirmed causes, resolution steps, and evidence of recovery.
- FR-10.2 An incident may be resolved after recovery is verified even if its root cause is still unknown.
- FR-10.3 The system shall allow responders to record corrective and preventive actions.

FR-11: Postmortems and Action Items

Priority: Must

Source: Team proposal; validate with Scribe and Incident Commander; interviews.

- FR-11.1 Users shall be able to prepare a postmortem containing the summary, impact, findings, timeline, resolution, and follow-up actions.
- FR-11.2 The incident timeline shall be available while preparing the report.
- FR-11.3 Each action item shall have an owner, due date, and completion status.
- FR-11.4 **Could:** The system may generate a draft from the incident timeline.

FR-12: Dashboard and Reports

Priority: Must

Source: Team proposal; validate with Engineering Manager; interview and survey.

- FR-12.1 The dashboard shall show total, open, and resolved incidents, severity counts, and average response and resolution times.
- FR-12.2 Users shall be able to filter by date, severity, status, team, and service, and view each incident's owner and SLA indicators.
- FR-12.3 **Should:** Live updates may deliver saved changes without a manual refresh. The dashboard itself is in the initial scope; live updates are an extension.

FR-13: Audit Log

Priority: Must

Source: Team proposal; validate with Platform Administrator and Audit Function; interview and document analysis.

- FR-13.1 The system shall record incident changes, permission changes, and changes to service and escalation settings.
- FR-13.2 Each record shall identify the user or system actor, action, time, and affected item.
- FR-13.3 Existing audit entries shall not be editable through the application.

FR-14: Service Catalog

Priority: Must

Source: Team proposal; validate with Service Owner; interview.

- FR-14.1 Authorized users shall be able to add and update services, their owners, criticality levels, and response and resolution targets.
- FR-14.2 **Could:** Users may record service dependencies to support investigation.

FR-15: On-Call Schedules

Priority: Must

Source: Team proposal; validate with Resolver and Platform Administrator; interviews.

- FR-15.1 Authorized users shall be able to assign responders to on-call time slots using one configured project time zone.
- FR-15.2 The system shall select the responder scheduled at the incident time, or use a configured fallback if no assignment exists.
- FR-15.3 **Could:** Automatic rotations and temporary schedule overrides may be added.

FR-16: SLA Tracking

Priority: Must

Source: Team proposal; validate with Service Owner and Incident Commander; interviews.

- FR-16.1 The system shall maintain separate response and resolution clocks for each incident.
- FR-16.2 Both clocks shall start at incident creation. The response clock shall stop at the first accepted acknowledgement; the resolution clock shall stop at accepted resolution.
- FR-16.3 The system shall calculate deadlines from the service targets and incident severity, display remaining time, and indicate breaches.
- FR-16.4 The initial version shall count elapsed time continuously.
- FR-16.5 **Could:** SLA pauses and business-hours calendars may be added after their rules are agreed.

FR-17: Previous Incident Search

Priority: Must

Source: Team proposal; validate with Resolver; interview.

- FR-17.1 Responders shall be able to search previous incidents using keywords and filter by service or severity.
- FR-17.2 Results shall provide access to previous findings, fixes, and postmortems that the user is allowed to view.

FR-18: AI Investigation Help

Priority: Should

Source: Team proposal; validate with Resolver; interview.

- FR-18.1 If included, local AI shall suggest investigation steps using the incident details and available previous records.
- FR-18.2 Suggestions shall be labelled as AI-generated and identify the records used.
- FR-18.3 **Could:** Log and deployment analysis may be added when those data sources are available.
- FR-18.4 Users shall be able to continue manually when AI is unavailable.

FR-19: Review of Suggested Actions

Priority: Should; review is required if AI suggestions are included

Source: Team proposal; validate with Incident Commander and Resolver; interviews.

- FR-19.1 If AI suggestions are included, an authorized responder shall review them before acting. This control is required whenever that feature is included.
- FR-19.2 The system shall record whether a suggested action was accepted or rejected.
- FR-19.3 **Won't (initial version):** The system shall not automatically execute remediation.

FR-20: Deployment Records and DORA Reports

Priority: Should

Source: Team proposal; validate with Engineering Manager and External Systems; interview and API document analysis.

- FR-20.1 If included, the system shall store deployment details and allow users to link incidents to deployments.
- FR-20.2 Reports may include Deployment Frequency, Lead Time for Changes, Change Failure Rate, and Mean Time to Restore using agreed definitions and available data.
- FR-20.3 Missing input data shall be shown as unavailable rather than reported as zero.

FR-21: External Communication

Priority: Could

Source: Team proposal; validate with Communications Lead and Customer Support Lead; interviews.

- FR-21.1 If included, customer-facing updates shall be separate from internal incident notes.
- FR-21.2 Only authorized users shall approve and publish external updates.

FR-22: Automated Health Checks

Priority: Should

Source: Team proposal; validate with Service Owner and External Systems; interview and API document analysis.

- FR-22.1 If included, the system shall check configured service endpoints and raise an alert after a configured failure threshold.
- FR-22.2 Recovery shall update the matching alert.
- FR-22.3 Automatic incident resolution shall require an explicit recovery rule.

3. Non-Functional Requirements

Performance targets and capacity limits, including user count, stored incident volume, and alert rate, will be set once during elicitation and initial testing on the available hardware. Until then, these limits remain provisional; no uptime percentage is promised.

NFR-01: Performance

Source: Team proposal; validate with Resolver; interview and prototype measurements.

- NFR-01.1 Incident creation, status updates, and dashboard loading shall be measured on the project test setup.
- NFR-01.2 AI generation shall be measured separately from core operations.

NFR-02: Availability

Source: Team proposal; validate with Resolver and Platform Administrator; interviews.

- NFR-02.1 Users shall receive a clear error when a requested operation is unavailable or fails.
- NFR-02.2 Failed writes shall not appear successful.

NFR-03: Capacity

Source: Team proposal; validate with Platform Administrator; interview and prototype measurements.

- NFR-03.1 The initial capacity scope is one organization with multiple users and services.
- NFR-03.2 Capacity testing shall cover concurrent users, stored incident volume, and incoming alert rate on the project test setup.

NFR-04: Security

Source: Team proposal; validate with Platform Administrator; interview and document analysis.

- NFR-04.1 Passwords shall be stored as salted password hashes, and the server shall check permissions for protected operations.
- NFR-04.2 Passwords, API tokens, and webhook secrets shall not appear in logs.
- NFR-04.3 If deployed beyond local development, client-server communication shall use HTTPS.

NFR-05: Reliability

Source: Team proposal; validate with Platform Administrator; interview and restart tests.

- NFR-05.1 A write operation shall be reported as successful only after it has been saved in the database.
- NFR-05.2 Saved incident records and escalation deadlines shall remain available after an application restart, provided the database is intact.

NFR-06: Usability

Source: Team proposal; validate with Resolver; interview and workflow review.

- NFR-06.1 The incident list shall clearly show severity, status, owner, and SLA information.
- NFR-06.2 Forms shall identify missing or invalid fields. Severity shall use labels or symbols as well as colour.
- NFR-06.3 The team shall review incident creation and acknowledgement with users during testing; no measured usability result is claimed yet.

NFR-07: Maintainability

Source: Team proposal; validate with Engineering team; code and test review.

- **NFR-07.1** Authentication, incident handling, notifications, and reporting shall be separated into modules.
- **NFR-07.2** Status, SLA, and escalation rules shall be testable without the UI or AI service.

NFR-08: Auditability

Source: Team proposal; validate with Scribe and Audit Function; interview and document analysis.

- **NFR-08.1** Timeline and audit entries shall include an actor and timestamp.
- **NFR-08.2** Existing entries shall not be overwritten through the application; corrections shall refer to the original entry.
- **NFR-08.3** Automatic deletion and advanced retention management are outside the initial scope.

NFR-09: Browser Compatibility

Source: Team proposal; validate with Resolver; interview and browser tests.

- **NFR-09.1** Core workflows shall be checked on Chrome and Firefox on desktop.
- **NFR-09.2** The tested browser versions shall be recorded. Support for other browsers and mobile layouts can be added later.

NFR-10: Backup and Recovery

Source: Team proposal; validate with Platform Administrator; interview and restore test.

- **NFR-10.1** The project shall provide a documented manual database backup and restore procedure.
- **NFR-10.2** The procedure shall be checked by restoring a test backup. Automatic backup schedules and fixed recovery-time guarantees are outside the initial scope.

NFR-11: Safe Concurrent Updates

Source: Team proposal; validate with Resolver; interview and concurrent-update tests.

- **NFR-11.1** If users acknowledge an incident at the same time, only one initial acknowledgement shall be accepted.
- **NFR-11.2** A conflicting update shall be rejected with a message rather than silently overwriting an accepted change.

NFR-12: AI Failure Handling

Source: Project constraint; validate with Resolver and engineering team; interview and failure tests.

- **NFR-12.1** If AI is included, models shall run locally through Ollama and requests shall have a configurable timeout.
- **NFR-12.2** Invalid output or AI failure shall not block core incident functions. Structured output shall be checked before use.

NFR-13: Testing

Source: Project constraint; validate with Engineering team; test review.

- NFR-13.1 Status, SLA, and escalation decisions shall use deterministic rules, not LLM calculations.
- NFR-13.2 Automated tests shall cover the main paths and important failures. Tests involving AI shall use fixed sample responses in CI.
- NFR-13.3 Mutation testing shall be applied to selected core rules as part of course testing work. No coverage or mutation-score target is claimed here.

NFR-14: Live Updates

Source: Team proposal; validate with Resolver; interview and reconnect tests.

- NFR-14.1 For the live-update extension in FR-12.3, connected boards shall display saved changes without a manual refresh.
- NFR-14.2 If the connection drops, the interface shall indicate the problem and refresh incident data after reconnecting.

4. Domain Requirements

These rules define how incidents, alerts, response times, and recovery are interpreted. Rules for optional features apply only if those features are included.

DR-01: Severity Levels

Source: Team proposal; validate with Service Owner and Incident Commander; interviews.

Level	Meaning
SEV1	Critical service or business impact
SEV2	High impact needing urgent attention
SEV3	Moderate impact
SEV4	Low impact

- DR-01.1 Each level shall have documented examples and impact criteria.
- DR-01.2 Severity describes impact. Response priority may also consider the time left before an SLA deadline.

DR-02: Incident Responsibilities

Source: Team proposal; validate with Incident Commander; interview.

- DR-02.1 The Commander coordinates the response; Responders investigate and fix; the Scribe records events; the Communications Lead manages updates.
- DR-02.2 One person may hold more than one responsibility when policy allows it.
- DR-02.3 Holding an incident responsibility shall not automatically grant system administrator permissions.

DR-03: Incident Lifecycle

Source: Team proposal; validate with Incident Commander and Resolver; interviews.

- DR-03.1 The normal flow is OPEN → ACKNOWLEDGED → INVESTIGATING → IDENTIFIED → MONITORING → RESOLVED → CLOSED.
- DR-03.2 If recovery fails, MONITORING may return to INVESTIGATING.
- DR-03.3 If recovery is possible without a confirmed cause, INVESTIGATING may move to MONITORING.
- DR-03.4 Any automatic-resolution transition shall be explicitly listed in the approved transition rules. No unlisted transition is allowed.

DR-04: SLA Timing Rules

Source: Team proposal; validate with Service Owner; interview.

- DR-04.1 Response time runs from incident creation to the first accepted acknowledgement.
- DR-04.2 Resolution time runs from creation to accepted resolution.
- DR-04.3 Service and severity determine the targets. Acknowledgement stops the response clock, not the resolution clock.
- DR-04.4 The initial version counts elapsed time continuously. Pause rules will be defined if that extension is included.

DR-05: Affected Services

Source: Team proposal; validate with Service Owner; interview.

- DR-05.1 Each incident shall identify its primary affected service.
- DR-05.2 If service dependencies are recorded, they indicate possible impact relationships; they do not prove a shared root cause.

DR-06: Postmortem Policy

Source: Team proposal; validate with Incident Commander and Scribe; interviews.

- DR-06.1 A postmortem shall record impact, timeline, findings, recovery steps, and follow-up actions.
- DR-06.2 An unknown root cause shall be recorded as unknown rather than guessed.
- DR-06.3 Which incident severities require a postmortem before closure will be confirmed during elicitation.

DR-07: Learning from Incidents

Source: Team proposal; validate with Engineering Manager and Resolver; interviews.

- DR-07.1 Reviews shall focus on contributing conditions and corrective actions.
- DR-07.2 Management reports shall use team- or service-level trends by default, rather than individual performance rankings.

DR-08: Grouping Alerts

Source: Team proposal; validate with Resolver and External Systems; interview and API document analysis.

- DR-08.1 Repeated delivery of the same event is a duplicate. A different alert about the same underlying issue is a related alert.
- DR-08.2 Alerts shall not be grouped only because they belong to the same service.
- DR-08.3 Alerts from different services may share an incident if configured rules or an authorized responder establish a common issue.
- DR-08.4 Adding an alert shall not reset the incident's creation time or elapsed SLA time.

DR-09: Acknowledgement and Escalation

Source: Team proposal; validate with Resolver and Incident Commander; interviews.

- DR-09.1 Acknowledgement means that someone has accepted responsibility. It does not mean the problem is fixed.
- DR-09.2 Acknowledgement shall stop escalation intended to find a responder.
- DR-09.3 Resolution-SLA breach notifications may continue after acknowledgement.
- DR-09.4 Reaching the last escalation level shall not automatically acknowledge or resolve the incident.

DR-10: Previous Fixes and AI Suggestions

Source: Team proposal; validate with Resolver and Incident Commander; interviews.

- DR-10.1 A previous fix is supporting information, not a guarantee that it suits the current incident.
- DR-10.2 AI suggestions shall not independently change incident status, calculate SLA results, or execute remediation.

DR-11: Recovery and Closure

Source: Team proposal; validate with Incident Commander and Service Owner; interviews.

- DR-11.1 Recovery of one alert shall not resolve an incident while other required recovery conditions remain unmet.
- DR-11.2 RESOLVED means recovery has been verified. CLOSED means the required review and documentation are complete.
- DR-11.3 Follow-up action items may remain open after closure if they have owners and due dates.